THE LAWS OF SOFTWARE TEAMS

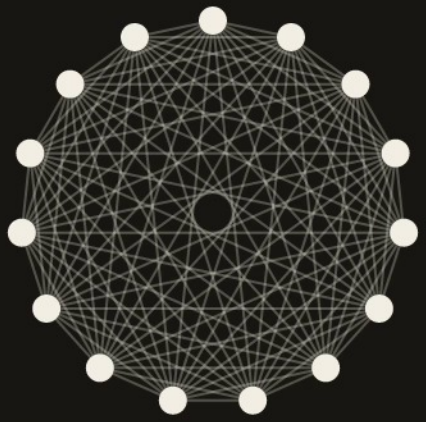
Dunbar's number

People can keep about 150 stable relationships

swipe →

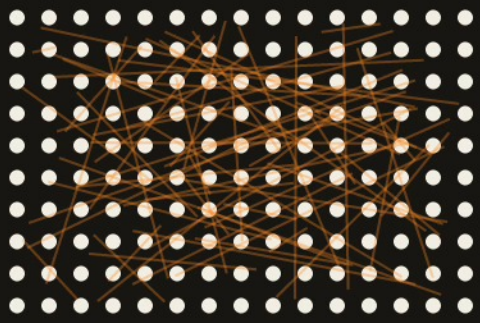
Ivan Rykov

At 15, everyone knows
At 150, nobody does



105

links · 15 people



11,175

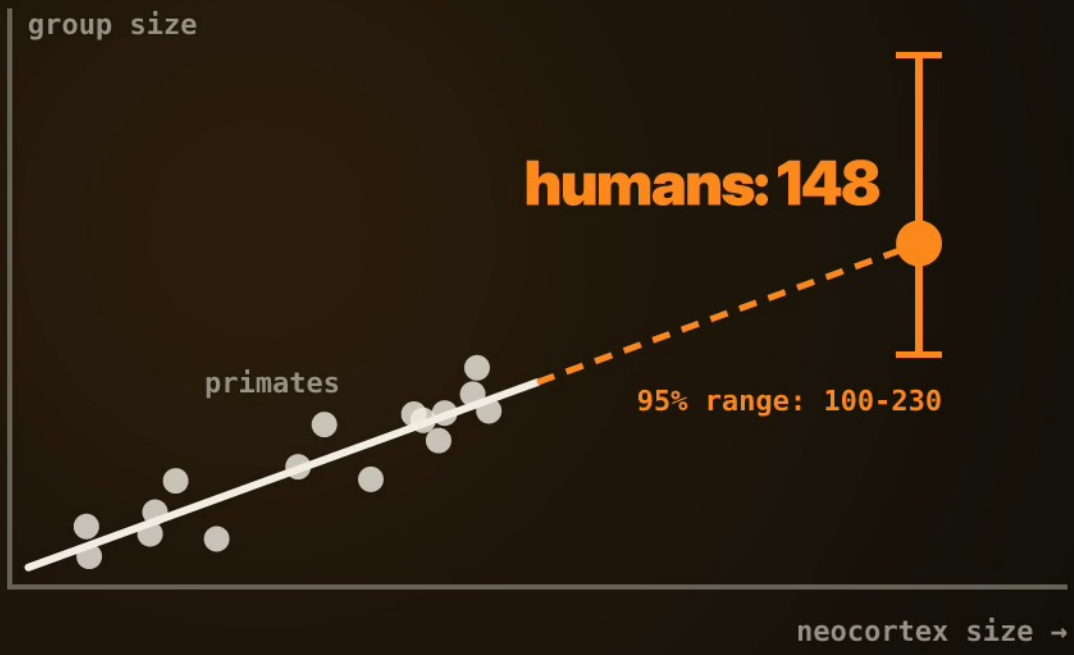
links · 150 people

`links between people = n × (n − 1) / 2`

There is a number for this

Dunbar's number: about 150

Robin Dunbar · anthropologist · 1992-93



Rounded to 150 stable relationships

Dunbar, 1992 and 1993 · primate points illustrative

Friends come in layers



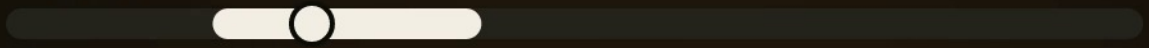
5 close · 15 best friends · 50 good friends · 150 friends

about ×3 per layer · then 500 and 1,500

Dunbar's layers · seen again in phone-call data (MacCarron et al., 2016)

A rough idea, not a law of nature

Dunbar, 1993 — 148 · range 100-230



Re-analysis, 2021 — 69 · range 4-292



other models: averages 16-109 · widest range 5-520

0 100 200 300 400 500

LINDENFORS, WARTEL & LIND · BIOLOGY LETTERS · 2021

"A cognitive limit on human group size cannot be derived in this manner"

Companies use it as a guide

W. L. GORE & ASSOCIATES

A new plant when one grew past about 150-200 people

Gladwell, The Tipping Point (2000): 150 · company histories: about 200

SPOTIFY · 2012

“Tribes are designed to be smaller than 100 people or so”

Kniberg & Ivarsson, Scaling Agile @ Spotify

Each layer needs a new way to talk



up to 10 people

a team · works by talking



~50 people

a team of teams · needs shared rituals

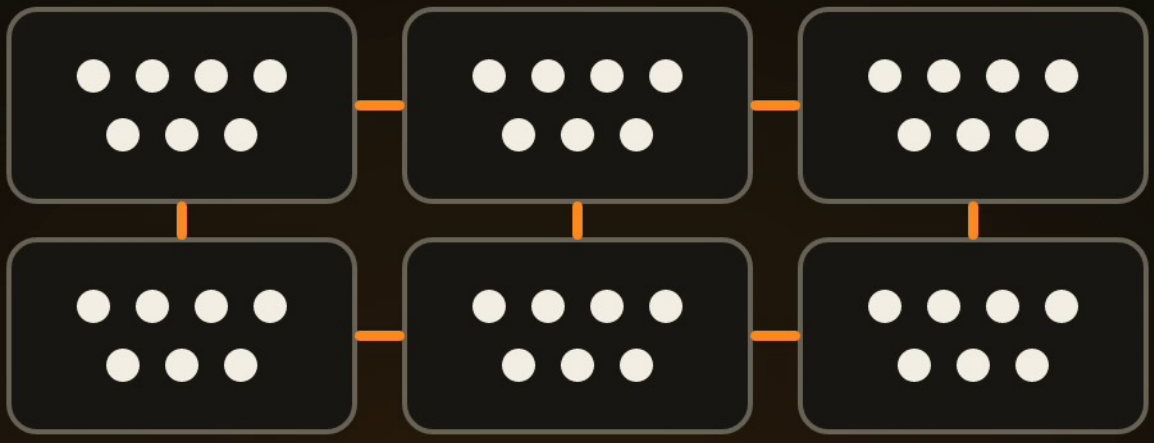


~150 people

a department · needs written decisions

a manager's reading of the layers, not Dunbar's research

Split early, along the system



Before the group stops working

CONWAY'S LAW

Team lines become system lines

illustration · amber: the interfaces between teams

08 TAKEAWAY

Grow in layers, not in one big room



Your org has a size limit, even if nobody wrote it down

WHAT I TAKE FROM THIS

So I add a new level of structure before the old one stops working:

1. **One group that works on one thing is ideally four engineers. A full team, with backend, frontend and QA, has up to ten people. If a lead also does technical work, they can manage up to ten people directly. A people manager can manage twenty at most**
2. **When a team grows, I split the team or the roles: one team lead who manages people, and separate tech leads, sometimes one for each area. I choose who joins which team by the priorities of the work, strengths and how people already work together**
3. **I split teams so that they overlap as little as possible. Some work stays shared, like the architecture and the test setup. It goes to chapters or guilds. Harder questions go to a technical council of a few strong engineers, so not everyone has to join**

WHAT I TAKE FROM THIS

4. **Between teams: demos every two weeks, write-ups that show what changes in each team, and a monthly talk by leadership about direction**
5. **I have 1:1s with my direct reports about every two weeks, and skip-level 1:1s about once a month or less often. Everyone should know they can go to someone higher who is neutral and can give advice**
6. **The first core engineers are hired slowly: often people I have worked with before, or after a very careful check that they share our values and approach. The team still needs different people. Someone drives, someone has ideas, someone does the hard daily work. Pushing back is useful, but not everyone should only push back. Later, hiring can go faster**

WHAT I TAKE FROM THIS

I have seen teams grow far beyond what the work needed. It was not about bad hires. More people do not replace clear goals and priorities. First the goals, then the size of the team

In the past, growing a team from 20 to 100 people needed managers for the new people, strict rules and processes. I think fewer products will need such teams. AI agents change the limits on team size: a small team can hand much of the coding to agents. But some limits stay. In my experience, one person can keep about five to seven parallel streams of work in mind. Someone also has to review a lot of generated code and judge it as a product, and AI is not yet reliable enough to do this alone. I think a product now needs a core of five to ten strong engineers

Sources: Robin Dunbar, 1992 and 1993 · Lindenfors, Wartel and Lind, 2021 · Dunbar's reply, The Conversation, 2021 · Zhou et al., 2005 · MacCarron et al., 2016 · Kniberg and Ivarsson, 2012

next: the Peter principle